

Chapter 82 — Tumours of the Penis

- **Uncommon but devastating**; diagnostically and therapeutically challenging for the urologist.
- **Rare in North America and Europe**; a substantial health burden in parts of Africa, South America and Asia.
- **Squamous cell carcinoma (SCC) is the commonest penile malignancy**; premalignant lesions are discussed first because they establish the anatomical, aetiological and histological relationship to SCC.
- **Prognosis and treatment planning rest on the pathological stage and histological features of the primary tumour and on the presence and extent of lymph node metastasis.**
- Contemporary themes: novel imaging, **dynamic sentinel node biopsy (DSNB)**, modified surgical approaches to improve staging accuracy and reduce morbidity, **organ-preserving surgery**, radiotherapy (primary and palliative), systemic and multimodal therapy, and a **risk-based scheme for the inguinal region** built on histological and clinical features.

PREMALIGNANT CUTANEOUS LESIONS

Penile Intraepithelial Neoplasia (Formerly Carcinoma in Situ)

Pathological and Clinical Correlates

- **Penile intraepithelial neoplasia (PeIN)**: dysplastic penile squamous epithelium with an **intact basement membrane**. Like invasive disease, divided into **human papillomavirus (HPV)-associated** and **HPV-independent (differentiated) PeIN**.
- **HPV-associated PeIN is the commonest form** and a precursor of invasive SCC; commonest subtypes **basaloid** and **warty**.
- **Mechanism of high-risk HPV (hrHPV) transformation**: viral **E6 binds p53**, viral **E7 binds retinoblastoma protein (Rb)** → cell-cycle progression and altered apoptosis.
- **E7 inactivation of Rb → overexpression of p16^{INK4a} (p16)** → p16 immunohistochemistry is the **surrogate marker of hrHPV** and the basis of routine classification.
- **HPV-independent (differentiated) PeIN**: hyperplastic squamous epithelium, keratin pearls, prominent intercellular bridges, atypical basal cells; **difficult to distinguish from reactive hyperplasia** (squamous or pseudoepitheliomatous hyperplasia).
- **2022 World Health Organization (WHO) classification**: grading PeIN I/II/III (mild/moderate/severe) is **discouraged** — **all PeIN is regarded as high grade**. "Low-/high-grade dysplasia" and "squamous carcinoma in situ" are to be **abandoned as pathological terms**.
- **Clinical eponyms** — erythroplasia of Queyrat (EQ, glans), Bowen disease (BD) and bowenoid papulosis (BP) (shaft) — are useful clinically but have **no unique histology** and are discouraged as pathological entities; all three fall under PeIN.

COMPARISON — CLINICAL ENTITIES OF PEIN

	Erythroplasia of Queyrat	Bowen disease	Bowenoid papulosis
Site	Glans (less often prepuce, uncircumcised)	Shaft	Shaft , young males
Appearance	Red, velvety , well-marginated; may ulcerate, discharge, pain	Sharply defined scaly erythematous plaques ; crusted or ulcerated variants	Papular (online)
HPV	HPV associated	—	HPV associated
Progression if untreated	10–33%	About 5%	Does not progress
Mimics	Chronic localised balanitis	BP, nummular eczema, psoriasis, superficial basal cell carcinoma	—

- **EQ histology:** atypical hyperplastic cells with disorientation, vacuolation, multiple hyperchromatic nuclei and **mitoses at all levels**; rete elongated, broadened, bulbous; submucosal capillary proliferation, ectasia and a **plasma-cell-rich infiltrate** — these distinguish EQ from chronic balanitis. Described by **Queyrat, 1911**.
- **Metastasis from PeIN is extremely rare** but reported.

Treatment of PeIN

- **Histological confirmation first:** multiple biopsies of adequate depth to exclude invasion.
- **Foreskin lesions:** **circumcision or excision with a negative margin** is adequate.
- **Glans lesions** are harder to excise while preserving anatomy → **glans resurfacing:** epithelium and subepithelial tissue dissected off the underlying spongiosum → defect closed with a **skin graft**; very low early local recurrence.
- **Alternatives:** topical **5-fluorouracil (5-FU)** cream, **5% imiquimod** cream, laser ablation (**neodymium:yttrium-aluminium-garnet [Nd:YAG]**, **potassium titanyl phosphate (KTP) 532 nm**, **carbon dioxide**).

KEY POINTS: VIRUS-RELATED DERMATOLOGICAL LESIONS

- **Male HPV infection** is associated with penile carcinoma and with **cervical dysplasia and carcinoma in female partners**.
- **Imiquimod cream** is the standard topical treatment of choice for **condyloma**.
- **Giant condyloma of Buschke-Löwenstein:** locally progressive, HPV related, **non-metastatic**, but needs **complete excision and follow-up** — coexisting SCC or malignant degeneration occurs.
- PeIN is an intraepithelial malignant process presenting as three clinical entities: **BP, BD, EQ**.
- **BP:** shaft of young males; **does not progress** to invasion.
- **Untreated progression:** BD about 5%; EQ up to 33%. **Metastasis rare**.
- **Goal:** cancer eradication with organ preservation.
- **Penile Kaposi sarcoma:** associated with **human herpesvirus 8**; prompts investigation for **human immunodeficiency virus (HIV)** or other immunosuppression.

Concept — PeIN is defined by an intact basement membrane, not by a grade: every lesion is high grade, the eponyms describe only site and appearance, and treatment is therefore organ-preserving eradication tailored to where the lesion sits.

SQUAMOUS CELL CARCINOMA

Epidemiology

- **0.4–0.6% of male malignancies** in the United States (US) and Europe; **up to 10%** in parts of Asia, Africa and South America.
- **Divergent trends:** falling in the US, India, Thailand, Philippines; **rising in several European countries** (including Denmark, the Netherlands, the United Kingdom, Scandinavia), China and Israel — attributed to **increasing HPV exposure and declining childhood circumcision**.
- **Global 2020 estimates:** age-standardised incidence about **0.8 per 100,000**, mortality about **0.3 per 100,000**.
- **Disease of older males:** abrupt rise in the **sixth decade**; mean age in the mid-50s. **Not unusual in younger males** — a meaningful minority are under 40, occasionally under 30; reported in children.
- **Surveillance, Epidemiology and End Results (SEER):** US incidence similar in White and Black males but **substantially higher in Puerto Rican Hispanic males**.
- **Racial and ethnic disparity in mortality:** independent predictors of worse survival are **higher stage, age over 65, African-American ethnicity and nodal disease**; Black patients carry about twice the disease-specific risk of death; US Hispanic, Black and Puerto Rican males have higher mortality than White males. **Causes unknown** — biology, access or treatment.

Aetiology and Risk Factors

- Incidence varies with **circumcision practice, hygiene, phimosis, number of sexual partners, HPV, tobacco, obesity** and other factors.

Phimosis and Circumcision

- **Neonatal circumcision virtually eliminates invasive penile carcinoma** by removing the **closed preputial environment**.
- **Smegma** (bacterial by-product of desquamated cells) is implicated through chronic irritation; **not proven carcinogenic** in humans, but the association is consistently observed.
- **Cycle:** poor hygiene → smegma → inflammation → **fibrotic healing** → **phimosis** → perpetuation.
- **Phimosis** (circumferential preputial fibrosis preventing retraction) is present in **25–75%** of patients and carries a **seven- to tenfold risk**.
- **The risk of non-circumcision is mediated by phimosis:** uncircumcised males with phimosis are at markedly raised risk; **without phimosis the risk is not significantly elevated** (underpowered).
- **Rare in Jewish populations** (universal neonatal circumcision); **less than 1% of US male malignancies**; **10–20%** in non-circumcising African tribes and Asian cultures.
- **Timing matters:** rare after neonatal circumcision, more frequent when delayed to puberty; **adult circumcision offers little or no protection** — the critical exposure has occurred by puberty/adulthood (atypia is already present in phimotic foreskins).
- **Neonatal circumcision protects against invasive cancer but not against carcinoma in situ (CIS);** phimosis associates with invasive disease but not CIS.
- **Circumcision protects against HIV-1; no protection** against herpes simplex virus type 2, syphilis or gonorrhoea.

Human Papillomavirus Infection

- **First clue:** wives or ex-wives of penile cancer patients had a **threefold risk of cervical carcinoma**; male partners of females with cervical intraepithelial neoplasia had more PeIN and more HPV.
- **Polymerase chain reaction (PCR) and in situ hybridisation** detect HPV DNA in penile lesions (benign and malignant) but **not in normal foreskin**. More than 40 types infect genital sites.

Low-risk: HPV 6 and 11

Genital warts; non-dysplastic lesions; also found in **non-metastatic verrucous carcinoma**.

High-risk: HPV 16, 18, 31, 33, 45, 52, 58

In situ and invasive carcinoma; **HPV 16 is the commonest** in primary tumours and is found in **metastases**.

- **E6 complexes TP53; E7 binds Rb** — via the **p14^{ARF}/MDM2/p53** and **p16^{INK4a}/cyclin D/Rb** pathways — with **p16 upregulation** as the immunohistochemical surrogate of hrHPV.
- **HPV infection correlates with number of lifetime sexual partners**, which also tracks penile cancer risk.
- **HPV is not invariable**: prevalence in penile cancer is **about half** → other pathways exist → classification into **HPV-associated and HPV-independent SCC** (2022 WHO).
- **HPV positivity varies by subtype**: **high in basaloid and warty, low in keratinising (usual) and verrucous**.
- **HPV-independent molecular events**: **CDKN2A silencing by promoter hypermethylation**, **INK4a/ARF-targeting** gene expression, **TP53 and p14^{ARF} mutations**, **MDM2 overexpression**; somatic variants (amplifications, deletions, loss of heterozygosity) and epigenetic change.
- **Inflammation-driven carcinogenesis**: lichen sclerosus (LS), phimosis and chronic balanitis → **reactive oxygen and nitrogen species** → DNA damage and genomic instability; **cyclo-oxygenase-2 (COX-2)** induction → prostaglandins and thromboxanes → proliferation, angiogenesis, invasion.

HIV Infection

- **HIV is a risk factor for HPV infection** (higher prevalence, **multiple genotypes**); both HPV and HIV are lower in circumcised males (African data).
- **HIV worsens disease**: about **fourfold risk of death** from penile cancer; progression from intraepithelial neoplasia occurs **about 6 years earlier**.

Tobacco Products

- **Smoking raises risk, dose-dependently**, independent of phimosis.
- **All forms — cigarettes, chewing tobacco, snuff — are independently associated**.
- Hypothesis: tobacco acts **with HPV or inflammation-associated bacteria**; shared with other anogenital carcinomas.

Penile Trauma and Lichen Sclerosus

- **Trauma**: carcinoma in the scarred shaft after **mutating circumcision** is a distinct entity; penile tears and rashes raise risk; **penile injury preceding onset** carries a very high odds ratio.
- **Lichen sclerosus (balanitis xerotica obliterans)**: subsequent cancer in **2.3–9%** on long follow-up; LS associates specifically with **HPV-independent carcinoma**.

Obesity

- **Risk factor for invasive cancer** via **buried penis**, inability to examine and clean → smegma and recurrent inflammation (**functional phimosis**); risk rises with each 5-unit increase in body mass index (BMI).
- **No convincing link** with occupation, other venereal diseases (gonorrhoea, syphilis, herpes), marijuana or alcohol.

Prevention

- **Neonatal circumcision is controversial**. American Academy of Pediatrics (AAP) evolution: no medical benefit (1989) → "potential medical benefits" (1999) → **August 2012: benefits outweigh risks**, justifying access for families who choose it (urinary tract infection, penile cancer, sexually transmitted infections including HIV).
- **Penile carcinoma is the only neoplasm with a predictable, simple prophylaxis that spares the organ at risk**.

- **Hygiene may substitute where good:** Denmark (circumcision rate very low) saw falling incidence as bathing facilities spread — but **rates have since risen, probably HPV-driven**, so circumcision may still matter in developed countries.
- **Modifiable prevention:** avoid phimosis (hygiene), **avoid HPV, avoid tobacco**.
- **HPV vaccination** should have an emerging role: bivalent (16/18, Cervarix), quadrivalent (16/18/6/11, Gardasil), **nonavalent Gardasil 9** (16/18/6/11/31/33/45/52/58); **efficacy proven in HPV-negative young individuals**; wider uptake is the goal.

Natural History

- **Begins as a small lesion** → extends over glans, shaft and corpora; papillary/exophytic or flat/ulcerative; **autoamputation** if untreated.
- **Flat ulcerative tumours** grow at a similar rate to papillary ones but have **earlier nodal metastasis and poorer 5-year survival**.
- **Lesions over 5 cm or covering more than 75% of the shaft** associate with more metastases and lower survival — not consistently reproduced.
- **Buck fascia is a temporary barrier**; penetration of Buck fascia and tunica albuginea → corporal invasion → **potential vascular dissemination**.
- **Earliest dissemination: regional femoral and iliac nodes**.

Lymphatic drainage — sequence

- Prepuce and shaft skin** → **superficial inguinal nodes** (external to fascia lata); **glans and corpora** → collar at penile base → superficial nodes → **deep inguinal nodes** (deep to fascia lata) → **pelvic nodes** (external iliac, internal iliac, obturator).
- **Consistent stepwise pattern** on lymphangiography; **cross-connections at every level** → **drainage is bilateral**.
 - **Nodal enlargement** → skin necrosis, chronic infection, death from inanition, sepsis or **femoral vessel erosion haemorrhage**.
 - **Clinically detectable distant metastases** (lung, liver, bone, brain) are uncommon (**1–10%**), **late**, usually after local treatment; **distant disease without nodal disease is unusual**.
 - **Relentless course:** most untreated patients die **within 2 years**; **no spontaneous remission** reported; **second primary neoplasms in 5–15%**.

KEY POINTS: EPIDEMIOLOGY, AETIOLOGY AND PREVENTION

- **Rare in developed countries**; varies with age, circumcision, HPV exposure, lifestyle and hygiene.
- **Disparity** in US incidence and outcome, notably for **Puerto Rican Hispanic males**.
- **Risk factors:** no neonatal circumcision, phimosis, HPV and HIV, tobacco, LS, obesity, trauma.
- **Largely preventable:** neonatal circumcision, HPV vaccination, behaviour modification.

Concept — *Two carcinogenic routes converge on the preputial sac: an HPV-driven route (E6/E7, p16-positive, basaloid/warty) and an inflammation-driven HPV-independent route (phimosis, smegma, LS, obesity) — which is why neonatal circumcision, hygiene and vaccination each prevent part of the disease, and adult circumcision prevents almost none.*

MODES OF PRESENTATION

Signs

- **The penile lesion itself usually alerts the patient:** subtle induration, small excrescence, papule, pustule, warty growth, luxuriant exophytic mass, shallow erosion, or **deeply excavated ulcer with rolled edges**.

- **Phimosis may conceal the tumour** → silent progression → eventually erosion through prepuce, **foul odour**, **discharge ± bleeding**.
- **Site distribution: glans commonest** (about half), then **prepuce**, glans plus prepuce, coronal sulcus; **shaft rare (under 2%)** — reflecting exposure of glans, sulcus and inner prepuce to irritants in the preputial sac.
- **Rare presentations:** inguinal mass, ulceration, suppuration or haemorrhage from nodes of a **tumour hidden under a phimotic foreskin**; urinary retention or urethral fistula.

Symptoms

- **Pain is disproportionately slight** and rarely the presenting complaint.
- **Weakness, weight loss, fatigue, malaise** from chronic suppuration; occasional significant **blood loss** from primary or nodes.
- **Symptoms of distant metastasis are rare at presentation** — local and regional disease is far advanced first.

Diagnosis

Delay

- **Patient delay is characteristic:** 15–50% wait more than a year — embarrassment, guilt, fear, ignorance, neglect — despite daily handling of the organ.
- **Physician delay:** prolonged antibiotic or antifungal courses before biopsy.
- Survival effect of delay is inconsistent across series, but **earlier diagnosis logically improves outcome**.

Examination

- **Most lesions are confined to the penis** at presentation.
- Assess the primary: **size, location, fixation, corporal involvement**; inspect penile base and scrotum.
- **Rectal and bimanual examination:** perineal body involvement, pelvic mass.
- **Careful bilateral inguinal palpation is extremely important.**

Biopsy

- **Biopsy confirming carcinoma, depth of invasion and grade is mandatory before definitive therapy** — it guides treatment of the primary and predicts occult nodal risk in cN0 patients.
- May be separate from definitive surgery; **dorsal slit** often needed for exposure; alternatively **biopsy with frozen section → immediate resection or ablation**.
- **Superficial biopsies mislead:** depth frequently indeterminable, grade discordant (especially verrucous and mixed patterns), occasional failure to detect well-differentiated cancer. **Adequate specimen cannot be overemphasised.**

Histological Features

- **Most penile tumours are SCC:** keratinisation, epithelial pearls, variable mitoses, disrupted rete pegs, basement membrane breached.

COMPARISON — HISTOLOGICAL SUBTYPES OF PENILE SCC BY HPV STATUS (WHO 2022)

Subtype	Share	HPV	Defining features	Behaviour / nodal risk
Usual type	45–65%	Independent	Typically grade II, invades spongiosum; LVI/PNI in about one-third	Nodes 25–40% ; behaviour set by size, grade, depth, LVI/PNI
Warty (condylomatous)	7–10%	Associated	Cauliflower-like; fibrovascular cores, papillomatous; jagged irregular border; grade II, bulky, slow	Nodes 17–25% ; vascular/perineural invasion uncommon
Basaloid	4–10%	Associated	Ulcerated irregular mass; small uniform basaloid cells, central necrosis, abundant mitoses, apoptosis; deeply invasive	High grade; over half node positive
Papillary	2–15%	Independent	Low grade; hyperkeratosis, papillomatosis; non-uniform papillae, no koilocytosis, jagged border (vs verrucous/warty)	Nodes uncommon (up to 12%)
Verrucous	3–7%	Independent	Low grade; broad pushing border ; straight papillae, very well differentiated; surface and interpapillary keratin	True verrucous: no metastatic potential ; recurs if under-excised
Sarcomatoid (spindle cell)	1–6%	Independent	SCC plus spindle component; mimics melanoma or sarcoma (leiomyo-, fibro-, angiosarcoma); deeply invasive, polypoid	Most aggressive ; vascular/perineural invasion, nodal and distant metastases
Rare (pseudoglandular, adenosquamous, clear cell, lymphoepithelioma-like)	≤1%	First two independent; last two associated	Deep corporal invasion	Aggressive; mortality highest in pseudoglandular and clear cell

- **Pattern:** HPV-associated subtypes split into a bulky low-risk (warty) and a highly aggressive (basaloid) form; HPV-independent subtypes span the spectrum from non-metastasising (verrucous) to the most lethal (sarcomatoid).

Grading

- **Broders classification** (1921, cutaneous SCC): keratinisation, pleomorphism, mitoses; four grades, often collapsed to three. **Low grade (I–II) is 70–80% at diagnosis.**
- **Low-grade lesions:** cords of atypical cells from a hyperkeratotic epidermis, keratin, intercellular bridges, pearls — absent in high grade.
- **Site-grade link:** almost half of shaft tumours are poorly differentiated, about 10% of preputial tumours → grade and stage correlate.
- **High grade correlates with nodal metastasis;** the unmet need is stratifying the majority low/intermediate-grade tumours.
- **Current: three-tier WHO/International Society of Urological Pathology (ISUP) grading:**
 - **Grade I:** resembles normal squamous tissue; irregular nests, little stroma.
 - **Grade III:** small irregular nests, poor keratinisation, pleomorphic hyperchromatic cells, frequent mitoses, marked stromal reaction.
 - **Any proportion of grade III (anaplastic) features = grade III;** the remainder = grade II.
- **Rationale: half of tumours are grade-heterogeneous, and any grade III component associated with nodal metastasis.**

Invasion variables

- **Vascular invasion predicted nodal metastasis in every study** assessing it.
- **Perineural invasion** (about one-third of cases) is a **strong predictor of nodal metastasis**.
- **Synoptic reporting** (College of American Pathologists protocol) is standard: core variables for **eighth-edition tumour-node-metastasis (TNM)** staging — grade, structure invaded, lymphovascular and perineural invasion, margins — plus optional variables.

KEY POINTS: NATURAL HISTORY AND PRESENTATION

- Begins on the **glans or prepuce** and enlarges progressively.
- **Delay** in presentation and in biopsy is common.
- **Examination of primary and groins** is critical to planning.
- Metastasis is by **lymphatic embolisation to inguinal nodes; distant metastases occur late**.

KEY POINTS: BIOPSY AND HISTOLOGICAL FEATURES

- **Adequate biopsy** is essential.
- Subtypes now fall into **two groups by HPV relationship**, with distinct morphology and behaviour.
- **Structure invaded, grade, vascular and perineural invasion** predict metastasis and define eighth-edition T stage.

Laboratory Studies

- **Often normal**; anaemia, leucocytosis, hypoalbuminaemia with chronic suppuration and malnutrition; **azotaemia** from urethral or ureteric obstruction.
- **Hypercalcaemia without bone metastases** (about one-fifth in one series): a function of **disease bulk**, linked to inguinal metastases, may **resolve after excision of involved nodes**; tumour production of **parathyroid hormone and related substances** → osteoclastic resorption.

Hypercalcaemia management: aggressive **saline hydration** → diuretic only if volume overloaded → **bisphosphonate first line** (pamidronate, etidronate, zoledronic acid; safer than mithramycin) → for severe neurological hypercalcaemia add a calciuric agent such as **calcitonin** for rapid lowering.

Radiological Studies**Primary penile tumour**

- **Physical examination misstages about one-quarter** (both under- and overstaging).
- **Ultrasound** (7.5 MHz linear): cancer is **invariably hypoechoic**; underestimates thickness; **cannot separate glans T1 from spongiosal T2**; but tunica albuginea always seen → **high sensitivity for cavernosal invasion**.
- **Magnetic resonance imaging (MRI)** with gadolinium: good soft-tissue detail; **not useful for T1**. **Computed tomography (CT)**: poor soft-tissue resolution, **not useful for the primary**.
- **Artificial erection** (intracorporeal prostaglandin E1) may augment contrast MRI; meta-analysis: **MRI specificity for corporal invasion is high with or without artificial erection**.

COMPARISON — STAGING THE PRIMARY FOR CORPORAL INVASION

Modality	Strength	Weakness	Role
Physical examination	Reliable for corporal invasion; most reliable for small distal lesions	Misstages some	Default
Ultrasound	Sees tunica albuginea	Underestimates depth; T1 vs T2 not separable	Adjunct
MRI ± artificial erection	Best soft-tissue detail ; high specificity for corpora	Not useful for T1	When examination is equivocal and organ sparing is planned
CT	—	Poor soft-tissue resolution	Not for the primary

- **Bottom line:** in the direct comparison, **examination was reliable for corporal invasion**; imaging adds value mainly when examination cannot be done properly. Small glanular lesions gain nothing from imaging.

Inguinal and pelvic region — clinically node-negative (cN0)

- **Examination reliability varies** with primary grade and stage and with body habitus.
- **CT and MRI depend on node enlargement** and cannot read internal architecture of normal-sized nodes; **CT** is usually chosen for groins, pelvis and distant disease.
- **CT and lymphangiography add nothing to examination** in impalpable groins (low sensitivity, high specificity).
- **Caveat:** CT helps in **obese patients and after prior inguinal surgery**, where examination is unreliable.
- **Fluorodeoxyglucose positron emission tomography/CT (FDG-PET/CT):** SCC is FDG-avid, but **sensitivity for occult cN0 disease is poor**.
- **Ultrasound-guided fine-needle aspiration cytology (FNAC)** in cN0 groins also has **limited sensitivity**.
- **No imaging modality detects microscopic nodal metastasis reliably.**

Clinically node-positive (cN+)

- **CT features — central necrosis or irregular nodal border —** are highly sensitive and specific for poor-prognosis pathology (three or more positive nodes, **extranodal extension [ENE]**, positive pelvic nodes).
- **FDG-PET/CT in cN+ or T3–T4:** good sensitivity and negative predictive value for **pelvic nodes**, high positive predictive value for **distant lesions**.
- **Distant metastases:** late, usually with significant inguinal/pelvic disease; commonest sites **lung, bone, liver**. For **bulky regional adenopathy:** chest, abdominal and pelvic CT, and **FDG-PET/CT performs best**.

cN0 groin

No imaging finds micrometastases → **surgical staging** decides.

cN+ groin

Imaging (CT features, FDG-PET/CT) **stratifies prognosis** and detects pelvic and distant disease.

KEY POINTS: RADIOLOGICAL IMAGING

- **MRI** gives the best soft-tissue detail of the primary; **examination** is most reliable for small distal lesions; MRI ± artificial erection helps when examination is equivocal.
- **Inguinal examination remains the clinical gold standard** in the non-obese patient; CT or MRI for the obese or previously operated groin.
- In proven inguinal metastasis, **CT abdomen/pelvis or FDG-PET/CT** identifies poor prognostic features and pelvic or distant disease unlikely to be cured by surgery alone.

Penile Cancer Staging

- Eighth edition American Joint Committee on Cancer (AJCC) / Union for International Cancer Control (UICC) TNM (2017) is used.
- **Grading:** three-tier WHO/ISUP; any proportion of anaplastic cells → grade III.

Definition of the primary tumour — changes from the seventh edition

- **Ta:** formerly "non-invasive verrucous carcinoma" (misread as covering all verrucous tumours, most of which are destructive with a smooth pushing front). Now: **pure verrucous carcinoma without overt destructive invasion**, and **non-invasive papillary, warty, basaloid or mixed carcinomas** — analogous to **pTa urothelial** tumours.
- **T1 anatomical detail:** "subepithelial connective tissue" replaced by **site-specific layers** (glans, foreskin, shaft).
- **T1a vs T1b validated:** pT1b has a markedly higher nodal rate than pT1a; **perineural invasion now joins LVI and high grade** in defining pT1b.
- **T2 vs T3 split: spongiosum (T2) vs cavernosa (T3)** — cavernosal invasion carries worse disease-specific survival and more nodal metastasis.
- **Urethral invasion removed** as a T3 criterion — no independent prognostic value.
- **T4** (scrotum, prostate, pubic bone): uncommon; may need major amputation, **neoadjuvant chemotherapy**, or palliative radiotherapy if unresectable.

Pathological N changes

- **pN1 widened to one or two unilateral nodes without ENE:** survival of two nodes equals one node (about 90% 3-year disease-specific survival) → more patients classed low risk, **avoiding adjuvant therapy recommended for pN2** (joint European Association of Urology–American Society of Clinical Oncology [EAU-ASCO] guideline).
- **pN2 = three or more unilateral or bilateral nodes:** clearly worse survival (about 60%).
- **pN3 = ENE or pelvic nodes:** ominous (about one-third 3-year survival); **adjuvant therapy strongly considered**.
- **Stage groups are driven by nodal status** — the eighth edition's strength is combining accurate primary staging with improved clinical and pathological nodal descriptors.
- T stage assigned by biopsy, **more reliably by complete resection**, incorporating grade, LVI/PNI and cavernosal involvement. **Palpable adenopathy plus primary histology determines further imaging;** FNAC of palpable nodes or of CT-detected pelvic nodes helps assign N before therapy.

TABLE 82.1 DEFINITION OF PRIMARY TUMOUR (EIGHTH EDITION AJCC TNM, WITH N, M AND STAGE GROUPS) — PETTAWAY ET AL., AJCC CANCER STAGING MANUAL, 2017

T	Criteria		
TX / T0	Cannot be assessed / no evidence of primary		
Tis	Carcinoma in situ (PeIN)		
Ta	Non-invasive localised SCC		
T1	Glans: lamina propria · Foreskin: dermis, lamina propria or dartos · Shaft: connective tissue between epidermis and corpora		
T1a	No LVI, no PNI, not high grade		
T1b	LVI and/or PNI, or high grade (grade III or sarcomatoid)		
T2	Corpus spongiosum (glans or ventral shaft) ± urethra		
T3	Corpora cavernosa (including tunica albuginea) ± urethra		
T4	Adjacent structures: scrotum, prostate, pubic bone		
N	Clinical (cN)	Pathological (pN)	
0	No palpable or visibly enlarged inguinal nodes	No nodal metastasis	
1	Palpable mobile unilateral inguinal node	≤2 unilateral inguinal metastases, no ENE	
2	Palpable mobile ≥2 unilateral or bilateral inguinal nodes	≥3 unilateral or bilateral metastases	
3	Fixed inguinal nodal mass or pelvic lymphadenopathy	ENE or pelvic nodal metastasis	
Stage group	T	N	M
0is / 0a	Tis / Ta	N0	M0
I	T1a	N0	M0
IIA	T1b, T2	N0	M0
IIB	T3	N0	M0
IIIA / IIIB	T1–3	N1 / N2	M0
IV	T4 any N; any T N3; any M1		

- **What the table tests:** T is defined by **structure invaded** (spongiosum vs cavernosa), with T1 split by **biology** (LVI, PNI, grade); N is defined by **number, laterality, ENE and pelvic spread**.
- **Traps:** T1b and T2 share stage IIA; any N3 or T4 is stage IV even without distant disease; urethral invasion no longer upstages; clinical N counts palpable nodes, pathological N counts metastases (pN1 allows two).

TABLE 82.2 EIGHTH EDITION AJCC: SUMMARY OF PENILE CANCER STAGING REVISIONS TO THE SEVENTH EDITION — PETTAWAY ET AL., 2017

Element	Change	Evidence level
Grade	Three-tier WHO/ISUP; any anaplasia = grade III	III
Ta	Broadened to non-invasive localised SCC	II
T1a/T1b	Perineural invasion added as discriminator	III
T1 site	Glans, foreskin, shaft with site-specific layers	I
T2 / T3	T2 = spongiosum; T3 = cavernosa	II
pN1 / pN2	pN1 ≤2 unilateral, no ENE; pN2 ≥3 unilateral or bilateral	II

- **Pattern:** every revision moves prognostically distinct groups apart (spongiosum vs cavernosa; T1a vs T1b) or merges prognostically identical ones (one vs two nodes).

TABLE 82.3 CARCINOMA OF PENIS: SUGGESTED DIAGNOSTIC PROCEDURES TO ASSIGN TNM STAGE

Category	Procedures
T	Examination; incisional–excisional biopsy (or complete resection) for grade, structure invaded, vascular and perineural invasion
N	Examination; CT if palpable (also if obese or previous groin surgery); FDG-PET/CT for bulky nodes; DSNB or modified/minimally invasive inguinal dissection for high grade, vascular/perineural invasion or invasive pattern; aspiration cytology as indicated
M	Examination; liver function and calcium ; CT chest, abdomen, pelvis; bone scintigraphy; or FDG-PET/CT as indicated

- **What it teaches:** T is a pathological stage (tissue required); N in the impalpable groin is **surgical**, not radiological; calcium belongs in the metastatic work-up.

KEY POINTS: STAGING

- **Presence and extent of nodal involvement determine survival** and must be recorded.
- The eighth-edition unified TNM is a consensus system with prognostic clinical and pathological descriptors.

Differential Diagnosis

- **Condyloma acuminatum, Buschke-Löwenstein tumour, balanitis xerotica obliterans.**
- **Infections:** chancre, chancroid, herpes, lymphogranuloma venereum, granuloma inguinale, tuberculosis — identified by skin tests, tissue studies, serology, culture or special stains.

Concept — The primary tumour is staged by tissue and the groin by surgery: examination and biopsy define T, imaging earns its place only when examination fails (obesity, prior surgery, equivocal corpora, bulky nodes), and the eighth edition is built so that nodal burden, not the primary, sets the stage group.

SURGICAL MANAGEMENT OF THE PRIMARY TUMOUR**Penile Preservation**

- **Amputation is the oncological gold standard** for rapid definitive control: local recurrence **0–8%**.
- **Cost:** reduced sexual quality of life; negative well-being and **psychiatric symptoms in up to half**. Relevant because **more than half of patients are 60 or younger**.
- **Favourable histology — Tis, Ta, T1; grade I–II** — carries lower metastatic risk and suits **organ- or glans-sparing** procedures.
- **Goals:** preserve **glans sensation** where possible, otherwise **maximise shaft length**.
- **Options:** topical 5-FU or imiquimod (**Tis only**), radiotherapy, Mohs surgery, limited excision, laser ablation.

Penile-Preserving Strategies**Circumcision and Other Excision Strategies**

- Circumcision, limited glans excision, and glans removal with shaft sparing preserve function and length. **Historical recurrence 11–50%** — but grade, size, location and margins were usually unreported.
- **Intraoperative frozen section** makes conservative surgery safe in selected discrete tumours.
- **The 2-cm margin dogma is abandoned:**
 - Microscopic proximal spread tracks grade — **5 mm for grade I–II, 10 mm for grade III; no skip lesions**.
 - Wide local excision or glans removal with margins well under 2 cm: **low local recurrence**; larger organ-sparing series: modest recurrence that **did not affect disease-specific survival**.

- Only margins **under 1 mm** raised local recurrence → **negative margin required; distance should exceed 1 mm** (small, low-grade tumours with negative frozen section).
- **Limitations:** deeply invasive proximal or distal tumours, **high grade**, and patients too unfit for salvage.
- In a large multinational glansectomy cohort, in **pT3 and high-grade** tumours local relapse was more frequent and **independently predicted worse cancer-specific and overall survival (OS)** → select **low-stage, low-grade** tumours.

Glans resurfacing (glans stripping) — for glans CIS

Document **absence of invasion** → subdermal dissection of skin and subepithelial connective tissue off the corpus spongiosum (**total** or partial) → **split-skin graft** → topical therapy for isolated CIS at a margin → careful follow-up.

- **Total resurfacing gives far fewer positive margins than partial;** some patients need re-excision for unexpected invasion at the margin; clinical recurrence rare.
- **Graft take high; erectile function preserved.**

Mohs Micrographic Surgery

- **Layer-by-layer excision with microscopic examination of each undersurface** (original **fixed-tissue technique**) → precise margin control with maximal preservation.
- Original series: high eradication; recurrence linked to **size of 3 cm or more**, advanced stage, prior failed therapy.
- **Not reproduced with modern frozen-section Mohs:** recurrence in about one-third (mostly salvaged by repeat Mohs) → **no clear advantage over excision with frozen-section margins.**

Laser Ablation

COMPARISON — LASER ENERGY SOURCES

Laser	Penetration	Consequence
Carbon dioxide	About 0.1 mm	Too superficial for CIS or T1; recurrence up to 50%
Nd:YAG (1064 nm)	Up to 6 mm protein denaturation	Preferred; with tumour-base biopsies , recurrence comparable to partial penectomy
Argon, KTP	—	Also used

- **Overall recurrence:** under 10% for CIS, 10–25% for T1 in older series; better with Nd:YAG plus base biopsies.
- **Autofluorescence and 5-aminolevulinic acid-induced fluorescence** proposed to target frozen-section biopsies.
- **High resumption of sexual activity and satisfaction.**
- Laser recurrence **equals excision for PeIN** but is **higher than excision for higher-stage tumours** → **PeIN is the optimal laser cohort.**

Contemporary Penile Amputation

- **Standard for deeply invasive or high-grade cancer:** partial or total penectomy.
- **Adverse features against organ preservation:** size 4 cm or more, grade III, deep cavernosal invasion.
- Organ preservation recurs more → **selection and follow-up compliance** are prerequisites; but **early-detected, treated recurrence does not compromise survival.**
- **EAU-ASCO:** discuss benefits and harms → **shared decision making.** Organ preservation for **PeIN, Ta, T1, grade I–II** in compliant patients.

TABLE 82.4 TREATMENT OF THE PRIMARY PENILE TUMOUR

Stage / site	Treatment
PeIN, glans	Laser, glans resurfacing; alternative topical
Ta, PeIN — foreskin or shaft skin	Excision to negative margin; alternatives laser, topical (Tis only)
Ta, T1 grade I–III — glans	By size, position, side effects: excision, resurfacing, glansectomy , radiotherapy (not for Ta)
Ta, T1 — foreskin or shaft	Complete excision to negative margin
T2, glans, no gross cavernosal involvement	Total glansectomy ± distal corporal transection; partial penectomy; radiotherapy
T2 with corporal invasion, T3	Partial or total penectomy
T4	Neoadjuvant chemotherapy then surgical consolidation in responders if resectability is in doubt
Local recurrence after conservative therapy	Excision to negative margin; may need partial/total penectomy; superficial low-grade recurrence may be re-treated conservatively
Radiotherapy	Selected T1–T2 of glans or coronal sulcus, under 4 cm

- **Pattern:** the treatment ladder follows **depth** — epithelium (topical/laser/resurfacing) → glans connective tissue and spongiosum (excision, glansectomy) → cavernosa (penectomy) → adjacent organs (chemotherapy first).
- **Traps:** topical therapy is for **Tis only**; radiotherapy is **not indicated for Ta**; T2 glans without cavernosal extension can still be glans-sparing.

KEY POINTS: SURGICAL TREATMENT OF THE PRIMARY TUMOUR

- Compliant patients with **small low-grade, low-stage lesions (PeIN, Ta, T1; grade I–II)** are optimal for organ preservation.
- Goals: keep **glans for sensation**, otherwise **maximise length**.
- Modalities: limited excision, Mohs, laser.
- Local recurrence is higher than with amputation, but **detected and treated early, survival is not affected**.
- **Amputation remains standard** for large, deeply invasive or high-grade tumours.
- In **pT3 or high-grade** tumours, local relapse is commoner and **reduces overall and cancer-specific survival**.

Concept — Margin width is a function of grade, not a fixed 2 cm: because low-grade tumours spread only millimetres and local recurrence in them is salvageable without survival cost, organ preservation is safe there — but in high-grade or cavernosal disease local relapse itself becomes lethal, and amputation regains its place.

TREATMENT OF THE INGUINAL NODES

Overview

- **Presence and extent of inguinal metastasis is the most important prognostic factor** — more than grade, gross appearance or microscopic pattern of the primary.
- **Lymphadenectomy alone can be curative**, unlike most genitourinary cancers — because penile SCC has a **prolonged locoregional phase** before distant spread.
- **Controversies:** selecting lymphadenectomy vs observation; low-morbidity staging procedures; multimodal therapy for bulky nodes. **No randomised trials** — retrospective and prospective series only.

Contemporary Indications for Inguinal Lymphadenectomy

Prognostic Significance of the Presence and Extent of Metastatic Disease

TABLE 82.5 CARCINOMA OF THE PENIS: PROGNOSTIC INDICATORS FOR SURVIVAL

Parameter	Range across series	Average / message
Patients with palpable nodes	About one-quarter to most	Varies with referral pattern
Clinically false positive (palpable, histology negative)	8–65%	Much palpable adenopathy is inflammatory
Clinically false negative (impalpable, histology positive)	0–66% (one outlier higher)	Occult metastasis is common in cN0
5-year survival, nodes negative	46–100%	About 73%
5-year survival, nodes resected and positive	0–86%	About 60% , driven by extent of nodal disease

- **What it is for:** node-positive disease is **curable by surgery**, but survival spreads widely because "node positive" hides differences in number, laterality, ENE and pelvic spread.
- **Trap:** examination alone neither confirms (inflammation) nor excludes (micrometastasis) nodal disease.

TABLE 82.6 FIVE-YEAR SURVIVAL (%) RELATED TO EXTENT OF NODAL METASTASIS — AND TABLE 82.7 FIVE-YEAR SURVIVAL RELATED TO PELVIC NODE METASTASES TREATED WITH SURGERY ALONE

Nodal burden	5-year survival
Minimal (≤ 2 positive nodes; in some series 1–3)	About 72–88%
Greater (>2 ; more than 3, 4–5, >5 , >6)	0–50% , falling with number
ENE in nodes 4 cm or larger	Very few 5-year survivors
Pelvic nodes positive, surgery alone (nine small series)	About 14% average

- **Criteria for about 80% 5-year survival after curative resection:** up to two involved nodes, unilateral, no ENE, no pelvic nodes.

Presence of Palpable Adenopathy as a Selection Factor

- **Palpable nodes are metastatic in about 43% on average;** the rest are inflammatory.
- **Persistent adenopathy after primary treatment** is usually metastatic; **new adenopathy during follow-up** is much more likely tumour.

Historical

4–6 weeks of **antibiotics** to separate inflammation from metastasis.

Current

FNAC of palpable nodes at or immediately after primary treatment — no 4–6-week delay; **antibiotic trial no longer advocated** (EAU-ASCO).

- **FNAC of palpable nodes:** high sensitivity and specificity.
- **Negative FNAC does not exclude** (false-negative 20–30% in older series) → close observation, **repeat aspiration or excisional biopsy** according to suspicion.

Evolving Indications for Lymphadenectomy Without Palpable Adenopathy

Immediate versus delayed surgery

- **Reluctance to routine inguinal lymph node dissection (ILND)** stems from morbidity: phlebitis, pulmonary embolism, wound infection, **flap necrosis**, **disabling lymphoedema** of scrotum and legs.

- **Morbidity reduced** by peri-operative care, technique, **myocutaneous flaps, preserving dermis, Scarpa fascia and saphenous vein**, reduced dissection extent, laparoscopic/robotic approaches.
- **Kroon et al. (2005), Netherlands Cancer Institute** — landmark: node-positive patients found by **early DSNB vs delayed dissection after surveillance: 3-year survival 84% vs 35%; ENE almost universal in the delayed group** vs uncommon in the early group.
- About seven series show **better survival with early vs delayed therapeutic dissection**; six show **delayed dissection rarely salvages** recurrence.
- In cN0 patients, inguinal relapse highest with observation, lower with DSNB, nil with lymphadenectomy; **initial DSNB independently reduced inguinal relapse and cancer-specific mortality** vs observation.
- → **Early surgical staging (ILND or DSNB) ensures intervention while tumour volume is small.**

Immediate / early staging

Small-volume, ENE-negative nodes; **curable**; lower complication rates.

Delayed therapeutic dissection

ENE usual; **salvage rare**; survival poor despite careful follow-up.

Morbidity versus benefit

- **Microscopic-disease dissection is less morbid than bulky-disease dissection** — less lymphatic tissue removed, venous drainage and skin blood supply preserved.
- **Operative mortality** (historically a few percent) is now near zero; deaths related to **sepsis**, with synchronous penectomy or palliative dissection → **patient selection, pre-operative antibiotics, wound care.**

TABLE 82.8 LYMPHADENECTOMY COMPLICATIONS IN FOUR SURGICAL SERIES — BEVAN-THOMAS, SLATON, PETTAWAY, J UROL 2002

Complication	Three historical series (1948–1990)	MD Anderson 1989–1998
Skin edge necrosis	About half or more	Markedly lower (significant)
Lymphoedema	About one-quarter to one-half	About one-quarter (significantly lower than the worst series)
Wound infection	About one-sixth	About one-tenth
Seroma	Under one-sixth	About one-tenth
Death	0 to about 1%	About 2%

- **What it shows:** the contemporary gain is concentrated in **skin necrosis**; lymphoedema remains the stubborn complication.

Fig. 82.3 Contemporary result: right ilioinguinal and left superficial dissection for T2N1M0 SCC; mild left oedema at 10 months; disease-free at 9 years.

Impact of primary tumour histology on predicting occult nodal metastasis

- Histopathological variables of the primary **stratify cN0 patients into lower- and higher-risk groups**, sparing the truly node-negative the morbidity of dissection.
- **PeIN and verrucous carcinoma: little or no risk** — metastasis from PeIN almost unknown; true verrucous tumours do not metastasise → **PeIN and Ta are low risk.**
- **Corporal invasion (seventh-edition pT2, now pT2–pT3): high risk** — average about 60% nodal metastasis, **similar whether or not nodes are palpable.**
- **T1 (subepithelial connective tissue only):** historically 4–14% nodal metastasis; exceptions with high rates reflect **grade and vascular invasion modifying stage.**
- **T1 grade I–II:** low pooled rate (about 7%); but **T1 grade II is heterogeneous** (about 11% pooled; 5–20% by thickness and growth pattern in the first nomogram) → prior EAU guidance placed **T1 G2 at intermediate risk.**
- **Vascular invasion:** the only multivariate predictor (venous and lymphatic embolisation) in one series; absent in all node-negative T1 tumours in another; in corporal-invasive tumours node positivity roughly **triples when vascular invasion is present.**

- **Perineural invasion** and the **microscopic front of invasion** independently stratify risk; PNI is now in the TNM.
- In a large multicentre pT1 G2 cohort, nodal risk low overall, **rising several-fold with LVI or PNI, and lowest when both absent**.
- **Nomograms** (stage, grade, LVI, p53) have **not been externally validated** and are not used.
- **Synthesis: pathological stage, grade, LVI and PNI drive the staging decision in cN0** — captured by TNM: **PeIN, Ta, T1a = low risk (about 10–18%); T1b or above = high risk (about one-third to over half)**.

TABLE 82.9 PENILE CARCINOMA: CORPOREAL INVASION AND INCIDENCE OF LYMPH NODE METASTASIS — AND TABLE 82.10 INCIDENCE OF NODAL METASTASIS FOR STAGE T1, GRADE I AND GRADE II PRIMARY TUMOURS

Primary tumour group	Nodal metastasis	Note
Corporal invasion (seven series; cN0 or unspecified)	About 40–90%; average about 60%	Same with or without palpable nodes
T1, grade I–II (five series)	Pooled about 7%	Low risk
T1, grade II (five series, largest multicentre)	Pooled about 11% ; single small series up to 44%	Node-positive tumours carried LVI or PNI

- **Pattern:** depth (corporal invasion) sets a high baseline; within T1 the **grade II group is where LVI and PNI decide the risk**.
- **Trap:** T1 G2 is not uniformly low risk — "grade II" without LVI/PNI is T1a, with either it is T1b.

Molecular prognostic markers

- **HPV/p16 status is prognostic:** hrHPV and **p16 expression correlate with better disease-specific survival**; p16 immunohistochemistry is routine, correlates with hrHPV, and is part of **standard reporting**.
- **p53:** normally blocks cell-cycle progression; **overexpression independently predicted nodal metastasis** in two studies.
- **Other nodal predictors:** superoxide dismutase 2 overexpression, **Ki-67 high, diffuse programmed death-ligand 1 (PD-L1)**, ID1 overexpression, raised **C-reactive protein (CRP)**, raised **neutrophil-to-lymphocyte ratio (NLR)** and **SCC antigen**.
- **Nomogram** of grade, LVI, PD-L1 and NLR: excellent concordance, **awaiting validation**.
- **Progression pathways:** **epithelial–mesenchymal transition** (vimentin, matrix metalloproteinases up, E-cadherin lost), CD44, **mammalian target of rapamycin (mTOR)**.
- **Barriers:** unstandardised methods, no large annotated tissue banks → **prospective multi-institutional validation needed**.

TABLE 82.11 PROGNOSTIC MOLECULAR MARKERS OF LYMPH NODE STATUS AND SURVIVAL IN PENILE CANCER: CURRENT STATUS — REVISED FROM MUNEE ET AL., WORLD J UROL 2009

Marker	Role	Nodal status	Survival
HPV / p16 ^{INK4a}	High-risk types disrupt TP53 and Rb; p16 rises with hrHPV	Contradictory / not associated	Better survival , less recurrence
TP53	Altered: proliferation, apoptosis, dedifferentiation	More metastasis (preliminary)	Worse
PD-L1	Inhibits T-cell function	More metastasis	Worse
SCC antigen (TA-4)	Serum marker, function unknown	Grossly evident metastasis	No role
Ki-67	Proliferation	Higher risk	No role
NLR	Inflammation	High → nodal metastasis	Worse
CRP	Inflammation	High → nodal metastasis	Worse

- **Pattern:** HPV/p16 is the only **favourable** marker and predicts **survival**, **not nodal status**; the rest are adverse.
- **Trap:** p16 positivity does not tell you the groin is negative.

Evolving Indications for Expectant Management of the Inguinal Region

- **PeIN, Ta, T1a:** low nodal incidence (0–18%) → **optimal for surveillance**.
- **T1b or greater:** one-third to over half node positive → **inguinal staging warranted**.
- **Non-compliant patients with invasive tumours** should be offered staging rather than observation.
- **Only about one-third of initially cN0 patients who later develop inguinal recurrence survive 5 years** → strict follow-up and **immediate intervention on any change**.

TABLE 82.12 PENILE CARCINOMA: SUGGESTED FOLLOW-UP FOR PATIENTS WITH NO INGUINAL ADENOPATHY WHO DO NOT UNDERGO INITIAL LYMPHADENECTOMY

Year	Low risk (Tis, Ta, T1a)	High risk (T1b or above)
1–2	3-monthly	2-monthly
3	6-monthly	4-monthly
4	6-monthly	6-monthly
5 onwards	Annually	Annually

- **Pattern:** intensity front-loaded into **years 1–2**, when inguinal recurrence occurs; the two schedules converge by year 4.

Indications for Modified and Traditional Inguinal Procedures

Modified Procedures

- In **cN0 patients selected by adverse primary pathology**, the goal is to establish whether metastasis exists with **minimal morbidity**. Options: FNAC, node biopsy, sentinel node biopsy, extended sentinel dissection, DSNB, superficial dissection, modified complete dissection.

Fine-needle aspiration cytology

- **Impalpable nodes** (lymphangiography- or ultrasound-guided): **limited sensitivity**.
- **Palpable nodes:** easy, **sensitivity up to 93%**, and a positive result immediately directs treatment.

Sentinel lymph node biopsy and extended sentinel lymph node dissection

- **Cabanas:** lymphangiography showed consistent drainage to a sentinel node **superomedial to the saphenofemoral junction**, near the **superficial epigastric vein**; if negative, no other node positive; if positive → complete superficial and deep dissection.
- **False negatives and non-sentinel metastases** reported → **anatomically directed biopsy is unreliable and no longer recommended**.

Dynamic sentinel node biopsy

- **Precise sentinel localisation with the lowest morbidity** of any surgical staging: **vital blue dye or fluorescein** plus **hand-held gamma probe** intraoperatively.
- **Netherlands Cancer Institute evolution:** initial false-negative rate about 18% → modifications:
 - **Pre-operative inguinal ultrasound ± FNAC** (subtle architectural change can divert lymph flow; positive FNAC → immediate dissection, bypassing DSNB).
 - **Serial sectioning and cytokeratin immunohistochemistry** of sentinel nodes.
 - **Exploration of groins with low or no signal**.
- **Two-centre update (Netherlands Cancer Institute and St George's, London):** false-negative rate **about 7%**. **Systematic review:** false-negative **3.5–22%**; minor complications **1–20%**, **major 0–7%**.
- **Guidelines:** at **high-volume centres with a standard protocol**, sensitivity is acceptable. **National Comprehensive Cancer Network (NCCN)** and **EAU-ASCO** recommend DSNB; **EAU-ASCO** names it the **preferred option**.

Superficial and fascia-sparing complete inguinal dissection

- **Superficial ILND (SILND):** removes nodes **superficial to fascia lata**; if frozen section positive → complete ipsilateral ilioinguinal dissection. **Rationale:** **deep nodes are not involved unless superficial nodes are**; negative SILND → no recurrence in follow-up → **no pelvic dissection needed** if negative.
- **Fascia-sparing ILND (FS-ILND):** superficial boundaries plus removal of nodes and fascia from the femoral vessels **within the fossa ovalis**; excellent disease-free survival in pN0–pN1, good in pN2 with adjuvant therapy.
- **Disadvantage:** **higher per-groin complications than DSNB** (about a quarter vs about one-seventh).
- **Advantages:** more information than a single node; **no risk of missing the sentinel** (all first-echelon nodes removed); **no specialised equipment**.
- Guideline-endorsed **alternatives when DSNB is unavailable or not preferred**.

Minimally invasive inguinal lymphadenectomy

- **Laparoscopic and robotic:** longer operating time, **equivalent nodal yield and recurrence, markedly fewer complications** (especially skin) than open.
- **Evidence almost entirely in cN0;** oncological results in higher-stage disease await validation. **EAU-ASCO:** acceptable for high-risk cN0 when DSNB is not feasible or preferred.

Radical inguinal and ilioinguinal lymphadenectomy

- In **resectable metastatic adenopathy**, therapeutic value justifies morbidity. Open complications: **wound infection, skin necrosis, lymphoedema, lymphocele, lymphorrhoea**.
- **Goals:** eradicate all gross cancer, **cover exposed vessels**, rapid healing (primary closure or myocutaneous flap).
- **EAU-ASCO:** cN1 → FS-ILND or radical ILND (R-ILND); cN2 or more → R-ILND. **Spare the saphenous vein** whenever oncologically feasible.

TABLE 82.13 RADICAL (R-ILND), FASCIA-SPARING (FS-ILND) AND SUPERFICIAL (SILND) INGUINAL LYMPH NODE DISSECTIONS — MODIFIED FROM BROUWER ET AL., EAU-ASCO GUIDELINE

	R-ILND	FS-ILND	SILND
Superior	External ring to anterior superior iliac spine (all)		
Lateral	ASIS to 20 cm inferior	ASIS to 20 cm inferior	ASIS to 17 cm inferior
Medial	15 cm down from pubic tubercle	15 cm	12 cm
Inferior	Apex of femoral triangle	Apex of femoral triangle	2–3 cm above apex
Floor	Sartorius, adductor longus muscles, femoral vessels	Sartorius and adductor longus fascia, femoral vessels	Muscle fascia and fascia over femoral vessels
Fascia preserved	No	On muscles; fossa ovalis fascia and deep nodes taken	On muscles and over femoral vessels
Saphenous spared	No	Yes	Yes
Sartorius transposed	Yes	No	No

- **Pattern:** the three differ at the **floor** — R-ILND goes to muscle and vessels; FS-ILND stays on muscle fascia but clears the fossa ovalis; SILND leaves all fascia and stops short of the triangle apex.
- **Trap:** FS-ILND removes deep nodes **within the fossa ovalis**; SILND does not.

COMPARISON — INGUINAL STAGING AND TREATMENT PROCEDURES

Procedure	Setting	Morbidity	Status
DSNB (with ultrasound ± FNAC)	High-risk cN0	Lowest	Preferred (EAU-ASCO) at experienced centres
SILND / FS-ILND	High-risk cN0; FS-ILND also cN1	Intermediate	Alternative; any experienced surgeon
Laparoscopic / robotic ILND	High-risk cN0	Lower than open	Acceptable alternative; cN+ unvalidated
Anatomical sentinel biopsy (Cabanas)	—	Low	Abandoned — unreliable
R-ILND	cN1 (option), cN2+	Highest	Standard therapeutic operation

Unilateral versus bilateral dissection

- **Lymphatic crossover → bilateral drainage is the rule.** The contralateral side may be limited to a **superficial dissection** if frozen section is negative.

Pelvic lymphadenectomy

- **No pelvic skip metastases** when inguinal nodes are negative.
- **Low pelvic risk with one or two small positive inguinal nodes**, no ENE, size under 3.5 cm → **pelvic lymphadenectomy (PLND) can be avoided.**
- **Positive pelvic nodes:** 5-year disease-specific survival (DSS) far worse (under one-fifth vs about 60%); about 14% across series → **PLND has little proven therapeutic value** but is an effective **staging tool** to select adjuvant chemotherapy or radiotherapy.
- **Pelvic metastases proven before surgery** → **neoadjuvant chemotherapy** then surgery in responders.
- **Boundaries:** **common iliac bifurcation** (superior), **Cloquet's node** (inferior), **genitofemoral nerve** (lateral), **bladder** (medial).
- **Distribution of positive pelvic nodes:** **external iliac commonest**, then obturator; common and internal iliac uncommon; **presacral none.**

- Open or minimally invasive; **open preferred for clinically positive nodes.**

Minimising Complications and Perioperative Care

- **Complication rate tracks the amount of nodal tissue removed:**
 - **R-ILND:** overall **most patients**; commonest surgical site infection, skin/flap necrosis, seroma, dehiscence, lymphoedema.
 - **SILND / FS-ILND:** intermediate; skin necrosis, seroma, lymphoedema.
 - **Laparoscopic / robotic:** about a quarter to a third; infection, seroma, lymphocele, dehiscence.
 - **DSNB:** major complications **under 7%.**

TABLE 82.14 IMPACT OF PRE-, INTRA- AND POSTOPERATIVE FACTORS ON COMPLICATIONS AFTER INGUINAL LYMPH NODE SURGERY — MODIFIED FROM BROUWER ET AL., EAU-ASCO GUIDELINE

Decrease complications	Increase complications	Uncertain effect
Transverse linear incisions	S- or T-shaped incisions	Wound vacuum
Saphenous vein sparing	Date-controlled drain	dressings
Volume-controlled drain removal	removal	Fibrin glue
Minimally invasive techniques	Sartorius transposition	Advanced bipolar
Lymphoedema prevention	Increased BMI	device
Deep venous thrombosis prophylaxis	Increased disease burden	
Surgical site infection prevention (antibiotics, shaving, skin preparation)	Sarcopenia	

- **Pattern:** each harmful factor has its paired opposite — transverse vs S/T incision; drain by volume vs by date.
- **Trap:** sartorius transposition, a classic step of radical ILND, increases complications; vacuum dressings and fibrin glue are unproven.

Risk-Based Management of the Inguinal Region

- **Assumptions:** primary controlled, pathological T stage known, groins examined; CT abdomen/pelvis and chest imaging as indicated.

TNM Stages pN1, Ta, T1a — Low Risk (Fig. 82.4A)

Impalpable groins → **observe** (compliant patients). **Palpable node** → **node biopsy (FNAC or core ± ultrasound)** → *negative*: antibiotics or observation; if node persists or grows → **excisional biopsy** → *positive at any step*: **ipsilateral ILND (superficial + deep)** with **contralateral SILND on frozen section, FS-ILND or DSNB** (experienced centres) → **ipsilateral PLND if more than two positive nodes or ENE** → adjuvant therapy as indicated.

TNM Stages T1b or Greater — High Risk (Fig. 82.4B)

- **Consensus:** poorly differentiated, LVI, PNI or pT2 or above → **surgical staging even with normal groins.**

Impalpable groins → **bilateral DSNB** (or SILND, FS-ILND or minimally invasive ILND) → both negative: **observe**; either side positive (permanent sections, second sitting): **ipsilateral ILND**; PLND optional, by pathology (more than two nodes or ENE).

Unilateral mobile node under 4 cm → node biopsy → positive: **ipsilateral ILND or FS-ILND ± pelvic dissection (frozen section) + contralateral superficial dissection, FS-ILND or DSNB**; negative: excisional biopsy.

Bilateral mobile nodes under 4 cm → pre-operative needle biopsy (counselling on extent and timing) → negative: excisional biopsy → negative: **observe**; **positive**: either **bilateral ilioinguinal dissection then adjuvant therapy**, or **neoadjuvant chemotherapy then surgery**.

- **4 cm is the arbitrary cut-off for surgery as monotherapy** — nodes above 4 cm carry more ENE.
- **Adjuvant therapy considered** if more than two positive nodes, bilateral metastases, ENE or positive pelvic nodes.

Bulky Adenopathy and Fixed Nodal Metastasis (Fig. 82.4C)

- **Survival depends on complete eradication** — rarely achievable by any single modality; **surgery plus chemotherapy** shows benefit.
- **Expanded MD Anderson neoadjuvant series** (mostly paclitaxel, ifosfamide, cisplatin [TIP]): about two-thirds responded; **5-year survival stratified by response** — highest with complete/partial remission, intermediate with stable disease, lowest with progression; a minority rendered pN0; **long-term survival tied to chemotherapy response and favourable post-resection pathology**.
- **Response improves resectability and survival**; surgery in non-responders occasionally cures but usually precedes rapid locoregional or distant relapse.
- **Palliative groin dissection rarely palliates**; **endoluminal vascular stents** give transient protection against femoral erosion.
- **Chemoradiation for cisplatin-ineligible patients**: phase II **mitomycin + capecitabine + radiotherapy** in stage IV M0 → most responded on FDG-PET/CT, over one-third completely.
- **International Penile Advanced Cancer Trial (InPACT; EA 8134; NCT02305654)**: neoadjuvant chemotherapy → ILND vs neoadjuvant chemoradiotherapy → ILND in advanced regional disease; will give high-level evidence.

Fixed nodes, mobile nodes 4 cm or more, or pelvic nodes (after imaging) → cisplatin-eligible? Yes: **induction/neoadjuvant chemotherapy** → responsive or stable → **ilioinguinal resection** of all residual disease; progressive → clinical trial, palliative chemotherapy, radiation to affected side → further progression → hospice. No: **radiotherapy ± chemotherapy (chemoradiation)** → responsive or stable → ilioinguinal resection; progressive → clinical trial or palliation → hospice.

KEY POINTS: TREATMENT OF THE INGUINAL NODES

- **Presence and extent of inguinal metastasis determine survival.**
- **Persistent palpable adenopathy** → **ultrasound- or CT-guided biopsy**, then scenario-appropriate management.
- In impalpable groins, **risk from the primary (TNM)** selects DSNB, superficial/open or minimally invasive ILND, or close follow-up.
- **High cure**: ≤2 inguinal nodes, unilateral, no ENE, no pelvic nodes; higher volumes → adjuvant or neoadjuvant therapy.
- Morbidity has fallen: fewer radical resections, better peri-operative care.
- **SILND, FS-ILND and minimally invasive ILND** reliably detect micrometastasis without special facilities but are **more morbid than DSNB**.
- **Modified DSNB**: low morbidity, externally validated at high-volume centres, **preferred staging for cN0** where feasible.
- Laparoscopic/robotic ILND: open-equivalent yields in cN0; **not yet routine for cN+**.
- **FS-ILND and R-ILND** remain standard for palpable (cN+) disease.
- **PLND** when more than two inguinal nodes are positive or ENE is present.

Concept — *Because penile SCC lingers in the groin before it spreads, the node is both the prognosis and the cure: everything in this section trades the survival gain of operating on small-volume nodes early against the morbidity of operating on negative groins — and TNM risk plus DSNB is the device that lets the surgeon win both.*

RADIATION THERAPY

Radiation Therapy of the Penile Primary

- **Curative potential with preservation of form and function**; **surgical salvage remains curative** for recurrence → reasonable first strategy in selected patients.

- External-beam radiotherapy (EBRT) or **interstitial brachytherapy**.
- **Circumcision before radiotherapy is necessary**: exposes the lesion, allows surface infection to settle, **prevents preputial oedema and later phimosis**, and may debulk the tumour.

External-Beam Radiotherapy

- **Advantages**: widely available, homogeneous dose, no interventional skill required.
- **Contemporary dose**: **60 Gy in 25 fractions (5 weeks) to 74 Gy in 37 fractions (7.5 weeks)**; older series used 50–55 Gy.
- **Set-up**: supine; penis held **vertical in a bivalved wax or Perspex block** with a central cylindrical chamber; minimal air gaps; larger chamber if oedema develops.
- **Skin dose problem**: megavoltage beams **spare the skin**, but penile cancer is cutaneous → the tissue-equivalent block acts as **bolus**, delivering full dose to the surface.
- **Results**: 5-year local control 44–70%; penile preservation about 40–66%; **EAU-ASCO pooled 5-year recurrence-free rate about 55% → at least an even chance of avoiding amputation**; salvage by partial or total amputation in most failures; cause-specific survival 58–96% by T and N.
- **Predictors of better control**: **dose above 60 Gy, overall time 45 days or less, fraction size at least 2 Gy**. **Worse control**: T3, over 4 cm, high grade.
- → **Minimum tumour dose 66 Gy in 2-Gy fractions over 6.5 weeks**. **Hypofractionation (over 2 Gy per fraction) may increase toxicity**.
- **Used most** in patients unfit for surgery (age, comorbidity) or with **locoregionally advanced disease** (primary treated in continuity with both groins and pelvis); may be combined with **weekly cisplatin** radiosensitisation (as in head and neck and cervix).

Brachytherapy

- **Iridium-192** via a temporary **parallel array of interstitial needles** through and around the tumour.

COMPARISON — BRACHYTHERAPY DOSE-RATE MODES

Mode	Delivery	Typical schedule	Outcome
Low dose rate (LDR)	Manually loaded Ir-192 wire/seeds; 50–60 cGy/hour continuous	About 60 Gy over about 5 days	Best-established ; highest pooled control
Pulsed dose rate (PDR)	Automated afterloader; hourly pulse = LDR hourly dose	As LDR	Radiobiologically equivalent to LDR
High dose rate (HDR)	10 Ci Ir-192 afterloader; catheter scaffold	Twice daily 3–4.3 Gy, ≥6 hours apart; 45–51 Gy over 7–8 days	Similar to LDR/PDR; wider availability, convenience
Surface mould	Ir-192 in a mould or cylindrical applicator worn on the penis	LDR era: 60 Gy surface over 7–10 days; HDR: 40 Gy in 5 days, 4 Gy twice daily	Short-term data only

Low-dose-rate brachytherapy

- **Institut Gustave Roussy** (LDR and PDR, over four decades): local recurrence under one-fifth; **dose under 62 Gy → local failure; tumour over 4 cm → worse disease-free survival**.
- **Late toxicity drivers**: dose over 60 Gy and treatment volume over 25 cc → **meatal stenosis**; dose over 65 Gy and dose rate over 0.43 Gy/hour → **painful ulceration**.
- **Penile preservation at 5 years about 85–88%**, falling to about two-thirds at 10 years.
- **Local failure can be very late** — most within 2 years, but some at 3.5–7.5 years and up to 23 years; yet **salvage penectomy needed in only about 3%**.

- **Nodes still matter:** before 2002, no prophylactic dissection → half of moderately or poorly differentiated tumours recurred regionally or distantly. Now **nodal indications mirror primary surgery — all pT1b or above get surgical nodal staging; biopsy supplies grade and LVI** when the primary is not resected.
- **LDR local control appears superior to EBRT** (EAU-ASCO pooled 5-year relapse-free rate about 79%) — but **selection bias:** brachytherapy is chosen only for **T1–T2, glans-confined, under 4 cm**.
- **Wider needle spacing (12–18 mm) reduced local failure** (wider lateral margin); with this, size did not predict failure, though others found failure above 4 cm and with corporal invasion.

High-dose-rate interstitial brachytherapy

- Established in breast, prostate, and essential in cervix; technique as LDR with a catheter scaffold.
- Limited cumulative experience; **penile preservation about 70–86% at 5 years**. Largest mature series: median about 49 Gy after biopsy (36 Gy after gross excision), **5-year local control about three-quarters, preservation about 70% → comparable to LDR/PDR**.

Surface Mould Brachytherapy

- For **non-invasive or very superficial tumours:** rapid dose fall-off with depth.
- **Careful selection:** depth is hard to judge clinically or radiologically, and a **dose margin** (analogous to a surgical margin) is needed.
- **HDR revival: three-dimensional-printed** custom moulds or cylindrical applicators; sources set off the skin to limit skin dose while reaching depth; **suits recurrence after laser or topical therapy**.
- **Reaction peaks about 3 weeks, heals over 6–10 weeks;** small series only, no long-term data.

TABLE 82.15 SELECTED SERIES REPORTING LOCAL CONTROL, CANCER-SPECIFIC SURVIVAL, COMPLICATIONS AND PENILE PRESERVATION AFTER PRIMARY EBRT OR BRACHYTHERAPY

Modality	Dose	5-year local control	5-year cancer-specific survival	Late toxicity	Penile preservation
EBRT (8 series)	35–74 Gy; 1.8–3.5 Gy/fraction	44–70%	58–96%	Stenosis up to about 30%; necrosis under 10%	39–66%
LDR / PDR (about 15 series)	50–70 Gy	About 65–90%	72–91%	Necrosis/ulceration up to about 40%; stenosis up to about 50%	About 65–100%
HDR (6 series)	36–54 Gy, twice daily	About 73–100%	83–100%	Necrosis and stenosis low (about 0–10%)	65–100%

- **What it is for:** brachytherapy buys **better control and preservation** at the price of **more local soft-tissue toxicity**; EBRT preserves less but is used for larger, less selected tumours.
- **Trap:** the brachytherapy advantage is partly **selection** (small glans tumours).

Tolerance of Radiotherapy

Acute effects

- **Moist desquamation** expected, more extensive after EBRT (larger volume); **re-epithelialisation in 4–8 weeks;** saline soaks and hygiene.
- Intercourse when comfortable, with **water-based lubricant**.

Late effects

- **Commonest: meatal stenosis and soft-tissue ulceration.**
- **Ulceration:** text quotes 0–8% after EBRT and 0–26% after interstitial therapy; individual brachytherapy series in Table 82.15 report up to about 40% (definitions and grading differ). **Persistent ulcer raises recurrence concern** → biopsy may be needed, but radiation ulcers are **flat and superficial without raised or exophytic**

edge; most heal conservatively (slower in **diabetes** and originally deep tumours) → antibiotics, **vitamin E**, steroid creams → **hyperbaric oxygen** if resistant.

- **Meatal stenosis in 10–45%**: linked to high dose per fraction (EBRT) or **close needle spacing near the meatus** (brachytherapy); **late, 18–24 months**, heralded by a weak, deviated or split stream → **self-dilatation** with a meatal dilator; otherwise stricture needing formal dilatation, rarely urethroplasty.
- **Sexual quality of life**: effects modest — most sexually active patients remain active; **erections largely maintained**, though many notice altered glans sensitivity; erectile dysfunction mild.
- **Brachytherapy spares more shaft** → **better erectile preservation** than EBRT; being more conformal, its **salvage is more often partial than total penectomy**.
- **Multidisciplinary tertiary assessment** is essential; **extended follow-up and self-examination** — up to **one in five local recurrences occur after 5 years**, salvageable without survival cost.
- **Summary: T1–T2 under 4 cm with no or minimal extension beyond the coronal sulcus suit brachytherapy**; better control and preservation with faster delivery (**4–5 days vs 6–7 weeks**); choice also depends on team expertise.

EBRT

Available, homogeneous, needs bolus; 6–7 weeks; larger/advanced tumours or unfit patients; ~55% relapse-free; less erectile preservation.

Brachytherapy

Skill-dependent; 4–5 days; T1–T2 glans <4 cm; ~79% relapse-free; better preservation; more local ulceration/stenosis.

Radiation Treatment of Lymph Node Regions

- **Surgical staging, not "prophylactic" groin radiotherapy**, is recommended for high-risk cN0 (EAU-ASCO, NCCN) → treatment tailored to actual pathology.
- **Surgery for resectable adenopathy** also yields **node count and ENE**.

Inguinal Radiotherapy

- **Adjuvant radiotherapy for node-positive penile SCC is controversial: no level 1 evidence**; not recommended in prior guidelines; used by extrapolation from **head and neck, cervix, vulva and anal canal**.
- **Supporting data: National Cancer Database** stage III patients after ILND — **adjuvant radiotherapy improved OS**; multicentre series — **omitting adjuvant pelvic radiotherapy after positive PLND worsened OS and DSS**.
- **More than two positive nodes, negative pelvis**: adjuvant radiotherapy gave **longer OS than adjuvant chemotherapy**, with no in-field failures despite frequent ENE — at **54 Gy for ENE, 57–60 Gy for gross residual**.
- **Dose is critical**: at 50 Gy in-field failures were frequent and benefit confined to **ENE-negative** patients; in pN3, **in-field failure doubled at 50 Gy or less**. Genomic modelling suggests **higher doses** are needed for primary and nodes.
- **HPV-positive patients benefit** from adjuvant radiotherapy (OS gain maintained on propensity matching; **not seen with adjuvant chemotherapy**).

Pelvic Radiotherapy

- PLND recommended for high-risk groin pathology (N2–N3) but not practised everywhere; Danish **inguinal-pelvic chemoradiotherapy** (all ENE, most bilateral) gave encouraging median OS.
- **InPACT** randomises high-risk groin pathology to **chemoradiotherapy vs PLND then chemoradiotherapy**.

Radiation Combined With Sensitising Agent

- **HPV/p16 status may predict chemoradiotherapy responsiveness** (as in head and neck): node-positive **p16-positive** patients had much better locoregional control after post-LND chemoradiotherapy; **regardless of HPV**, adjuvant chemoradiotherapy improved 2-year locoregional control.
- Adjuvant chemoradiotherapy improved cancer-specific survival (CSS) at 1–2 years despite **worse selected patients** (more nodes, pN3, ENE) — **OS equalled that of more favourable untreated patients**.

- **InPACT**: about 200 node-positive patients (UK and North America) randomised to sequencing of surgery, neoadjuvant chemotherapy and chemoradiation.
- **Recommendation (from vulvar cancer)**: **adjuvant ipsilateral groin radiotherapy for two or more positive nodes or ENE**; dose at least 50 Gy/5 weeks, 54 Gy for ENE, 57–60 Gy for residual disease; include the pelvis unless pelvic nodes are known clear. HPV-positive patients may gain more.
- **Palliative radiotherapy** helps painful bone metastases and cord compression; a **short 5–10-fraction course is poor for bulky or fixed groin nodes** → consider **cisplatin plus radiotherapy** if glomerular filtration rate (GFR) over 50 mL/min.
- **Summary**: groin radiotherapy is **not prophylaxis** (unless unfit for surgery); it is **adjuvant for multiple nodes or ENE**; chemoradiotherapy may convert inoperable disease or be definitive.

Fig. 82.5 8 years after LDR brachytherapy (60 Gy over 5 days) for T2N1 SCC with concurrent node dissection: hypopigmented scar, minor telangiectasia. **Fig. 82.6** 5 years after HDR brachytherapy (38.4 Gy, 3.2 Gy twice daily for 6 days) after circumcision and excision with positive margins (T1, 8-mm invasion). **Fig. 82.7** Moist desquamation 3 weeks after HDR surface mould (40 Gy, 4 Gy twice daily) for CIS. **Fig. 82.8** Inoperable left inguinal nodes: **weekly cisplatin 40 mg/m²** with volumetric modulated arc therapy — 45 Gy/25 fractions to pelvic nodes, 63 Gy/35 fractions to left groin → PET-CT clear at 4 months; no leg oedema at 2 years.

KEY POINTS: RADIATION THERAPY

- Radiation is an effective **penile-preserving** option; **brachytherapy for T1–T2 under 4 cm**; EBRT also for larger or locoregionally advanced tumours ± sensitising chemotherapy.
- **One in five recurrences occurs after 5 years** → continued follow-up; salvage penectomy may cure.
- **Nodal staging criteria are identical** after primary radiotherapy or surgery.
- **pN2–N3** may benefit from adjuvant radiotherapy with **dose escalation** to ENE and residual disease.
- Unresectable nodes may become operable after **neoadjuvant chemotherapy or chemoradiation**.
- Integration being tested in **InPACT (EA 8134)**. Palliative radiotherapy helps metastatic disease.

Concept — Radiotherapy trades dose against tissue tolerance at a cutaneous target: skin must be fully dosed (bolus, circumcision), local control follows total dose and overall time, toxicity follows dose per fraction and volume — and none of it removes the need to stage the groin surgically.

SYSTEMIC THERAPY FOR PENILE CANCER

- **No completed randomised trials**; retrospective and phase II data show meaningful responses to **platinum-based** regimens.
- **Most active: triplets** — **TIP and docetaxel, cisplatin, 5-FU (TPF)**; doublets (cisplatin + 5-FU, paclitaxel + carboplatin) may be less toxic but lack prospective efficacy data.
- **Two figures for triplet survival**: the text quotes median OS of **7–14 months** for triplets; Table 82.16 gives **17.1 months for TIP**. The difference is the **setting** — the TIP figure comes from the **neoadjuvant N2–N3 M0** population, the lower range from locally advanced or metastatic disease.

TABLE 82.16 SAFETY AND EFFICACY OF MULTIDRUG PENILE CANCER REGIMENS WITHOUT BLEOMYCIN

Regimen (study)	Drugs and schedule	Setting	Response	Median OS	Treatment deaths
TIP — Pagliaro 2010 (landmark)	Paclitaxel 175 mg/m ² d1; ifosfamide 1200 mg/m ² d1–3; cisplatin 25 mg/m ² d1–3; q3w	Neoadjuvant, N2–N3 M0 (phase II)	50%	17.1 mo	None
TPF	Docetaxel 75 mg/m ² d1; cisplatin 60 mg/m ² d1; 5-FU 750 mg/m ² /day d1–5; q3w	Locally advanced/metastatic (phase II)	38.5%	13.9 mo	None; significant toxicity
Cisplatin + 5-FU (retrospective)	5-FU 800–1000 mg/m ² /day d1–4 infusion; cisplatin 70–80 mg/m ² d1; q3w	Retrospective	32%	8 mo	None
Irinotecan + cisplatin	Irinotecan 60 mg/m ² d1, 8, 15; cisplatin 80 mg/m ² d1; q4w	Phase II	About 31%	4.7 mo	None

- **What it shows: taxane-plus-cisplatin triplets** give the highest response rates; TIP is the only regimen with data in the neoadjuvant setting.
- **Trap:** the table deliberately excludes **bleomycin** — its toxicity is unacceptable.

Neoadjuvant Chemotherapy

- **Bulky nodes = extensive lymphatic disease;** few are cured by surgery alone. Neoadjuvant therapy treats systemic disease early and **downsizes nodes** → complete resection in responders.
- **Pagliaro et al. (2010)** — landmark phase II: **four cycles of neoadjuvant TIP for clinical N2–N3: objective response rate (ORR) 50%**, a few pathological complete responses, median time to progression about 8 months, **median OS 17.1 months**; long-term disease-free survival in about two-thirds of responders vs very few non-responders. → **Neoadjuvant TIP for cisplatin-eligible patients with high-risk inguinal nodes or any pelvic nodes.**
- **TPF:** phase II missed its primary end point with significant toxicity; neoadjuvant TPF gave poor progression-free survival and frequent discontinuation for toxicity → **may be too difficult for some.**
- **Meta-analysis of neoadjuvant studies:** pooled ORR about half, overall mortality about half. **No standard regimen; evidence favours cisplatin-plus-taxane.**
- **Summary:** neoadjuvant cisplatin-containing chemotherapy is active for **unresectable primaries or bulky nodes** in fit patients, enabling curative resection.

Adjuvant Chemotherapy

- **Historical Milan (Pizzocaro) VBM** — vincristine, bleomycin, methotrexate, 12 weekly courses in high-risk node-positive patients: very few relapses; adjuvant VBM roughly doubled 5-year survival vs surgery alone. **Unconfirmed and will not be re-studied** because of bleomycin and methotrexate toxicity.
- **Pelvic-node-positive retrospective series** (TIP, TPF, bleomycin–methotrexate–cisplatin, VBM): adjuvant chemotherapy associated with about a year's survival advantage and lower all-cause mortality — confounded; needs prospective confirmation.
- **National Cancer Database:** no survival difference after adjustment.
- **Bottom line:** **no strong evidence that adjuvant chemotherapy improves OS**; relapse-free survival poorly studied; benefit likely greatest in **pN3, especially pelvic node involvement** → balanced discussion.

Neoadjuvant chemotherapy

Phase II evidence (TIP); response predicts resectability and survival; **favoured** for bulky/fixed or pelvic nodes.

Adjuvant chemotherapy

Retrospective, conflicting; no proven OS gain; **consider in pN3 / pelvic nodes.**

Palliative Systemic Therapy

Chemotherapy

- **Platinum-based first line is preferred**; triplet vs doublet and cisplatin vs carboplatin chosen by **comorbidity**.
- **TPF and TIP each combine two neurotoxic drugs** → unsuitable with **pre-existing neuropathy or low GFR**.
Paclitaxel–carboplatin extrapolated from other SCC sites.
- First-line chemotherapy roughly **triples median OS versus best supportive care** (months, not years).
- **Vinflunine** single agent (prospective; **Eastern Cooperative Oncology Group [ECOG] performance status ≤2**, GFR ≥60 mL/min): modest response, median OS about 8 months, manageable toxicity.
- **Paclitaxel** second line: modest ORR, median OS about 5 months.
- **Second-line chemotherapy is minimally effective**: median OS **6 months or less**; single-agent vinflunine or taxane is an alternative to platinum but less effective.

Immunotherapy

- Checkpoint blockade, **HPV-directed vaccines**, adoptive T-cell therapy under study; rationale: **relatively high PD-L1 expression**.
- Checkpoint inhibitor trials: **low response rates (about one in six)**; basket trials (nivolumab + cabozantinib ± ipilimumab; ipilimumab + nivolumab) included few penile patients with occasional partial response or stable disease.
- **Routine checkpoint inhibition is not supported**; chemotherapy-exposed patients belong in **early-phase basket trials**.

Targeted Therapies

- Beyond first line, options are limited; **actionable alterations are rare**.
- **Epidermal growth factor receptor (EGFR)** targeting (as in other SCC): sporadic palliative responses to **cetuximab** and **panitumumab**; **dacomitinib** (oral pan-HER inhibitor) phase II: response in about one-third, one complete response.

COMPARISON — SYSTEMIC AGENTS BY CLASS AND LINE

Class	Agents	Target / mechanism	Line	Status
Platinum triplet	TIP , TPF	Cytotoxic	Neoadjuvant ; first-line palliative	Most active
Platinum doublet	Cisplatin + 5-FU; paclitaxel + carboplatin	Cytotoxic	First-line palliative (less fit)	Less toxic; limited data
Single agent	Vinflunine, paclitaxel	Microtubule	Platinum-unfit or second line	Less effective
Bleomycin regimens	VBM, BMP	Cytotoxic	Historical adjuvant	Discouraged — toxicity
Anti-EGFR	Cetuximab, panitumumab (antibodies); dacomitinib (oral pan-HER)	EGFR / HER family	Refractory	Sporadic activity; trials
Checkpoint inhibitors	Nivolumab, ipilimumab (± cabozantinib)	PD-1 / CTLA-4	Refractory	Not routine; trials

KEY POINTS: CHEMOTHERAPY

- **Neoadjuvant cisplatin-containing chemotherapy should be offered for bulky regional nodes** — response may allow curative resection; optimal multimodal strategy undefined without level 1 evidence.
- Patients unfit for or declining neoadjuvant chemotherapy may benefit from **chemoradiation**.
- **Bleomycin is discouraged** — unacceptable toxicity.
- **Surgical consolidation** for fit patients with objective response; consider in stable disease.
- **Progression on chemotherapy = poor prognosis** → trials or symptom-directed care.

Concept — *Systemic therapy in penile cancer earns its value by making surgery possible: response to neoadjuvant cisplatin-taxane chemotherapy is itself the prognostic selector for resection, whereas post-operative and later-line therapies have never shown a convincing survival gain.*

NONSQUAMOUS PENILE MALIGNANT NEOPLASMS

- **About 5% of penile neoplasms (SCC 95%)**; evidence from case reports and small series — incidence, pathology, treatment, and parallels to non-genital counterparts.
- **Melanoma and basal cell carcinoma are rare because penile skin is sun-protected.**

Basal Cell Carcinoma

- **Very rare** on the penis (tiny fraction of male BCC); **commonly proximal shaft near the base** — contrast with distal SCC.
- Slow growing, long diagnostic delay; **local excision virtually always curative**; Mohs used; metastasis essentially unreported (single case).
- **Fibroepithelioma of Pinkus**: premalignant variant on the shaft; diagnosed and cured by excision.
- No consistent link with immunosuppression, other skin cancers or prior radiation.
- Mesenchymal (smooth or striated muscle, fibrous, fatty, vascular) malignancies are also rare; treatment from case series.

Melanoma

- More than 150 cases; **under 1% of penile cancers**.
- **Blue-black or reddish-brown pigmented papule, plaque or ulcer on the glans** (prepuce less often). Histology: atypical junctional activity with pigmented cells displaced into the dermis.
- **Clark (level) and Breslow (thickness)** not prospectively validated on the penis; the **AJCC cutaneous melanoma system** (incorporating both) was used in the largest series.
- Older series: distant spread in most; contemporary: **early-stage node-negative tumours do well**.
- **Spread: haematogenous via corporal vasculature**; lymphatic permeation to inguinal and pelvic nodes.
- **Surgery is primary**; radiotherapy and chemotherapy adjunctive or palliative.
- **Historical**: partial or total penectomy with en bloc bilateral ilioinguinal dissection for localised or single-region disease.
- **MD Anderson algorithm**: foreskin → **circumcision**; glans → **partial penectomy**; glans-shaft → **partial or total**; **bilateral modified ILND if Breslow ≥ 1 mm, ulceration, or Clark IV–V**. DSNB unproven but reasonable at experienced centres.
- **Mucosal-site melanoma** (glans, meatus, fossa navicularis, distal urethra) looks more aggressive than cutaneous, possibly from diagnostic delay; **stage-matched outcomes are similar**.
- **Checkpoint immunotherapy has revolutionised** recurrent/metastatic melanoma — adjuvant **nivolumab** after resected stage IIIB–IV gave high 12-month recurrence-free survival → **melanoma multidisciplinary team** or expert referral.

Sarcomas

- Rare; benign and malignant mesenchymal tumours equally common; newborn to eighth decade; **same presentation for benign and malignant** — mass, pain, enlargement, **priapism**, obstruction; may **mimic a Peyronie plaque**.
- **Malignant: proximal shaft; benign: distal**. Commonest malignant: **vascular (haemangioendothelioma)**, then neural, myogenic, fibrous; case reports of malignant fibrous histiocytoma, angiosarcoma, leiomyosarcoma, epithelioid sarcoma, osteosarcoma.
- **Superficial** (integumentary) → wide local excision or partial amputation; **deep** (corporal) → total amputation. **Local recurrence is characteristic**.
- **Leiomyosarcoma: under 2 cm → local resection**; deeper → partial or total amputation; **deep lesions at the base have the worst prognosis**.
- **Regional and distant metastases rare → no node dissection unless palpable**; aggressive local treatment aims at cure. Radiotherapy and chemotherapy unassessed.

Extramammary Paget Disease

- Rare; large series from China and Korea. **Erythematous, eczematoid, well-demarcated** — clinically indistinguishable from EQ, BD or CIS; pruritus, discomfort, serosanguineous discharge; penis, scrotum, perianal region, axilla — **sites of apocrine glands**.
- **Paget cells**: large, clear, hydropic cells with hypochromatic nuclei; **cytokeratin 7, carcinoembryonic antigen (CEA), gross cystic disease fluid protein positive; S-100 negative**.
- Behaves as a **slow intraepithelial apocrine adenocarcinoma**; may invade and spread to nodes via dermal lymphatics.
- **Associated genitourinary malignancies** (prostate, bladder, kidney) and others — nearly half had another cancer, **usually diagnosed before** the EMPD → screen.
- **Treatment**: resect **skin and dermis with up to 3 cm gross margin**; frozen sections guide extent but **false negatives are common** without dermatopathology.
- **Outpatient mapping biopsies** (sequential peripheral biopsies until all negative) → **fewer positive margins and lower local recurrence**; positive margins → further resection; intraepidermal disease with negative margins did not recur. Cover with **local skin or scrotal flaps**.
- Deep invasion → extended resection; **inguinal adenopathy → radical node dissection, poor prognosis**. Chemotherapy (docetaxel–carboplatin neoadjuvant; cisplatin–epirubicin–paclitaxel for metastatic) shows promise. **Evidence-based multidisciplinary EMPD guideline** now exists (diagnosis, malignancy screening, intraepidermal and invasive disease, non-surgical options, metastatic disease, follow-up).

Adenosquamous Carcinoma

- Glandular plus squamous elements **independent of urethral glands**; large (5–9 cm) firm grey-white granular exophytic mass of distal shaft or glans; mucin- and CEA-positive glands.
- Case-level: local excision ± limited dissection with long survival; total penectomy with delayed ILND and PLND (pT2N3) disease-free at 5 years; one p16/hrHPV-positive case cured by **definitive chemoradiation**.

Lymphoreticular Malignant Neoplasm

- **"Primary" penile lymphoma is probably a misnomer** — usually an early sign of **systemic disease**; the penis has no lymphoid tissue → **mandatory search for systemic disease**.
- **Leukaemic corporal infiltration → priapism**.
- Commonest: **diffuse large B-cell lymphoma**; presents as penile lump or priapism; other sites nodes, lung, liver, brain.
- **Systemic therapy even if the penis is the only site** — commonly **rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone (R-CHOP)** — treating local disease, occult deposits, and preserving form and function.

Metastases

- Rare (a few hundred cases) despite rich vascular and lymphatic supply; **commonest sources: bladder, prostate, rectum**; also gastrointestinal tract, testis, kidney.
- **Routes:** direct extension, **retrograde venous and lymphatic** transport, arterial embolism.
- **Commonest sign: priapism** (replacement of one or both **corpora cavernosa**); also swelling, nodularity, ulceration, obstruction, haematuria; cutaneous, preputial or glanular deposits less common.
- **Differential:** idiopathic priapism, venereal or infective ulcers, tuberculosis, Peyronie plaque, primary tumours.
- **Signals advanced disease:** usually appears soon after diagnosis of the primary; **most die within a year** (median survival months). **Prostatic-origin** metastases (often **ductal features**) survive longer, treated with radiotherapy and androgen deprivation → **histology guides therapy and prognosis**.
- **Palliation:** excision by partial or total amputation for solitary or distal disease; **proximal corporal invasion → cure unlikely**; penectomy for intractable pain; **MRI** maps proximal extent and satellites; **dorsal nerve section** and suprapubic cystostomy for pain and drainage; systemic therapy for the primary histology, trials or hospice.

COMPARISON — NONSQUAMOUS PENILE MALIGNANCIES

Tumour	Typical site / sign	Spread	Treatment	Prognosis
BCC	Proximal shaft, near base	Almost never metastasises	Local excision / Mohs	Excellent
Melanoma	Glans, pigmented	Haematogenous (corpora) and lymphatic	Excision by site; ILND by Breslow/ulceration/Clark; immunotherapy	Stage-dependent
Sarcoma	Proximal shaft (malignant)	Local recurrence; metastasis rare	Wide excision or amputation; no routine node dissection	Worst for deep basal lesions
EMPD	Penoscrotal eczematoid plaque	Intraepidermal first; dermal lymphatics	Wide excision, mapping biopsies	Good if intraepidermal; invasive can be lethal
Lymphoma	Lump or priapism	Systemic	R-CHOP (systemic even if isolated)	Disease-dependent
Metastasis	Priapism	From bladder, prostate, rectum	Palliation; treat primary histology	Poor (months)

KEY POINTS: NONSQUAMOUS MALIGNANT NEOPLASMS

- **BCC:** highly curable, low metastatic potential.
- **Sarcomas:** prone to local recurrence; metastases rare; superficial lesions treated less radically.
- **Melanoma:** aggressive but curable early with appropriate surgery; immunotherapy and targeted therapy for advanced disease; multidisciplinary management.
- **EMPD:** intraepidermal spread first; **wide local excision to negative margins**; invasive EMPD can be lethal; guideline available.
- **Penile metastases:** usually from an obvious existing primary; poor prognosis; therapy directed to primary histology and local palliation.

Concept — Each nonsquamous tumour behaves like its counterpart elsewhere, modified by penile anatomy: sun protection keeps BCC and melanoma rare, corporal vascularity carries melanoma and metastases (hence priapism), apocrine glands explain EMPD, and absent lymphoid tissue makes penile lymphoma a systemic disease until proven otherwise.